RabeeQasem - ImpactAi at Daleel 2026: Conformer Based Arabic Argumentative Discourse Mining

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Abstract

We present our system for Daleel 2026 Task 2 on Arabic Argumentative Discourse Unit (ADU) detection. Our main contribution is a Conformer-based tagging architecture that combines a pretrained AraBERT encoder with a Conformer head to capture both global discourse context and local span-boundary cues. The system also uses overlapping token windows, class-weighted training, character-level span reconstruction, and lightweight post-processing. On the official hidden test set, our system achieved an overall F1 score of 0.732, substantially outperforming the official baseline of 0.285 and ranking fourth in the closed track.

1 Introduction

Argument mining aims to identify and classify the argumentative components that structure opinions, evidence, and reasoning in text (Nabhani et al., 2026),(Lawrence and Reed, 2019), (Stab and Gurevych, 2017). The Daleel 2026 shared task focuses on Arabic argumentative discourse mining, addressing an important gap in Arabic NLP, where discourse-level argument mining resources remain limited. The task covers two forms of Arabic argumentative discourse: news editorials and debates. In Task 2, systems are required to identify and classify Argumentative Discourse Units (ADUs) and determine their character-level boundaries. The task is evaluated using a modified F1 score that accounts for partial span overlap between gold and predicted annotations.

We participated in the closed track, which restricts systems to the organizer-provided training and development data, publicly available pretrained models, standard NLP tools, and rule-based or hybrid methods. Our main strategy formulates Task 2 as a sequence tagging problem using BIO labels over tokenized Arabic paragraphs (Ramshaw and Marcus, 1995). We fine-tuned an AraBERT-based encoder (Antoun et al., 2020) with a Conformer tagging head (Gulati et al., 2020) to capture both contextual evidence patterns and local boundary cues. Since argumentative spans often cover most of the paragraph, the system predicts token-level ADU labels and then maps them back to character-level spans. We further applied post-processing rules to improve span consistency, reduce fragmented predictions, and adjust selected label decisions based on development-set analysis.

Our system achieved 4th place in the closed-track leaderboard. our experiments show that Arabic ADU detection is highly sensitive to span boundaries, label imbalance, and confusion between semantically related evidence types such as Anecdote, Statistics, and Testimony. We also found that lightweight post-processing, especially merging fragmented spans and applying label-specific rules, improved the final submission without requiring additional training data. The code and resources used in our system are available at The code and resources used in our system are available on github.

2 Background

2.1 Task Setup

Daleel 2026 Task 2 is formulated as an Argumentative Discourse Unit (ADU) detection task for Arabic texts, covering two genres: news editorials and debate transcripts (Nabhani et al., 2026). Given an input paragraph, its identifier, and its genre, systems must identify all argumentative units by returning their exact character start and end offsets, along with one of six argumentation labels: Common Ground (CO), Assumption (AS), Testimony (TE), Statistics (ST), Anecdote (AN), and Other (OT).

For example, a training instance from a debate contains:

نتحدث الآن بنقطتين ومحورين أساسيين. المحور

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الأول نتحدث فيه عن الصراع الروسي الأوكراني،
ونقطتنا الأولى أن هذا التوسع جعل الدول
الأخرى في حالة تأهب للحرب التي تهدد الأمن
العالمي...

The gold annotation marks the opening organizational segment as OT, while subsequent spans are annotated as AS, corresponding to argumentative assumptions about expansion and alliances. Such examples illustrate a primary difficulty of the task: precise boundary detection. Argumentative units are often lengthy, adjacent to one another, and separated only by punctuation or short discourse markers.

2.2 Dataset Details

The training data consists of 612 paragraphs (255 editorials and 357 debates), the development set contains 217 paragraphs (88 editorials and 129 debates), and the final hidden test set contains 213 paragraphs. As shown in Figure 1, the training set exhibits severe class imbalance. The most frequent label is AS (1,739 spans), followed by OT (492 spans), AN (343 spans), and TE (313 spans). The rarest labels are CO and ST, with only 50 and 38 spans respectively. This imbalance makes the detection of fine-grained evidence types particularly challenging.

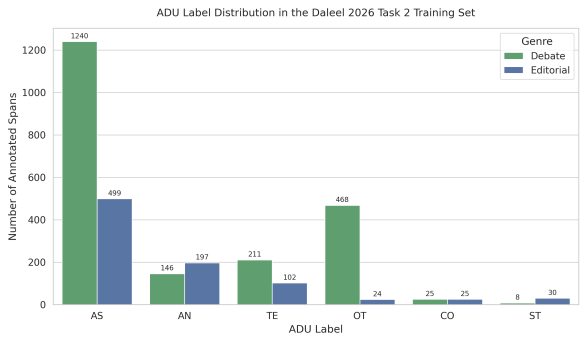


Figure 1: Distribution of ADU labels in the Daleel 2026 Task 2 training set across the two text genres: editorials and debates. The figure shows the strong class imbalance in the dataset, especially for CO and ST.

2.3 Related Work and Motivation

Argument mining has commonly been studied as a task that requires identifying argumentative components and assigning them discourse or argumentative roles. The Daleel label set is related to prior work on argumentation strategies in news editorials, where argumentative units are categorized according to the type of evidence or discourse function they provide, such as assumptions, testimony, statistics, anecdotes, and common ground . For Arabic NLP, pretrained transformer encoders such as AraBERT have shown strong performance across Arabic understanding and token-level tasks, making them a suitable backbone for Arabic ADU detection (Alatawi and Kadhem, 2025).

Our contribution focuses on adapting a Conformer-based architecture to Arabic argumentative discourse unit detection. The Conformer architecture was originally proposed for speech recognition, where it combines self-attention for global sequence context with convolution for local pattern modeling (Gulati et al., 2020). This combination is relevant to ADU detection because the task requires both paragraph-level contextual understanding and accurate local boundary recognition. As far as we know, our system is the first to apply a Conformer-based structure to Arabic argumentative discourse unit detection. This design allows the model to use attention for discourse-level evidence patterns while using convolution to capture local cues such as punctuation, discourse connectives, and short argumentative markers.

3 System Overview

Our system is designed as a span-level Arabic ADU detection model. The overall pipeline is shown in Figure 2. Given an Arabic paragraph, the text is tokenized using the AraBERT tokenizer and converted into token IDs, attention masks, and offset mappings. Long paragraphs are divided into overlapping token windows to handle the maximum input length of the encoder. The tokenized input is then passed through the AraBERT encoder, followed by a Conformer-based tagging head and a linear classification layer. Finally, token-level predictions are reconstructed into character-level ADU spans and refined using post-processing rules.

3.1 Model Architecture

Our model consists of three main modules: a pretrained AraBERT encoder, a Conformer tagging head, and a linear token-level classification layer. The AraBERT encoder provides contextualized representations of the input tokens, while the Conformer head further models both global con-

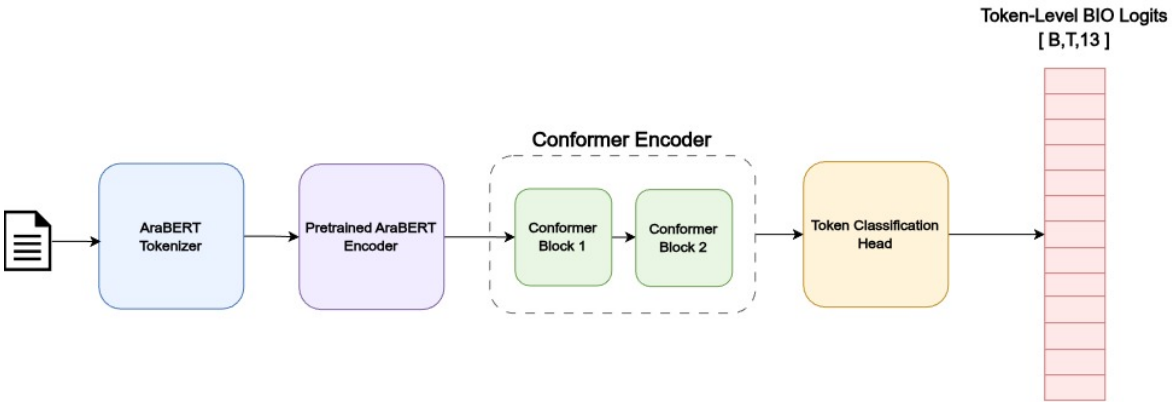


Figure 2: Overview of the proposed ADU detection pipeline. The system processes an Arabic paragraph using AraBERT tokenization, overlapping token windows, an AraBERT encoder, a Conformer tagging head, and a token-level classifier before reconstructing final character-level ADU spans.

textual dependencies and local boundary patterns. The resulting representations are then passed to the classification layer to produce token-level predictions for ADU detection and classification.

3.1.1 AraBERT Encoder

The first module is a pretrained AraBERT encoder. We use AraBERT to obtain contextualized Arabic token representations, which are important for distinguishing between argumentative functions that depend on sentence-level and paragraph-level context. Since the task is in Arabic and includes both editorials and debates, AraBERT provides a strong pretrained representation for Arabic text while remaining compatible with the closed-track setting.

3.1.2 Conformer Tagging Head

The second module is a Conformer-based tagging head composed of two Conformer blocks. The Conformer architecture combines self-attention and convolution, making it suitable for modelling both global context and local boundary patterns. In our task, self-attention helps capture discourse-level dependencies across the paragraph, while convolution helps detect local cues such as punctuation, connectives, and short argumentative markers. Figure 3 illustrates the Conformer block used in our system.

3.1.3 Token-Level Classification Layer

The final module is a linear classification layer that maps each token representation to an ADU tag. The classifier predicts whether each token belongs to one of the six ADU categories or falls outside an argumentative span. These token-level predictions are later converted into character-level spans using the tokenizer offset mappings.

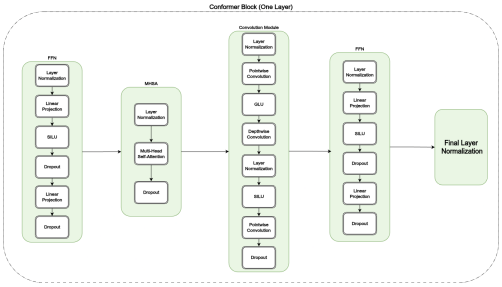


Figure 3: Structure of the Conformer block used in the tagging head. The block combines feed-forward layers, multi-head self-attention, and a convolution module to capture both global contextual information and local span-boundary cues.

3.2 Resources Used

The only external resource used in our system was the publicly available pretrained AraBERT model. We also used standard machine learning and NLP libraries, including PyTorch and Hugging Face Transformers (Wolf et al., 2020), alongside the official evaluation script released by the shared task organizers.

3.3 Addressing Task Challenges

The task presents several challenges: long paragraphs, class imbalance, span-boundary sensitivity, and confusion between semantically related labels. To process long paragraphs, similar to (Stefan and Nisioi, 2026) and (Wu et al., 2025), we split tokenized text into overlapping windows (max 512 tokens, stride 128) to preserve boundary context. We applied class-weighted loss during training to mitigate severe label imbalance

(especially for CO and ST). For precise boundary detection—which the official soft span-overlap F1 metric heavily rewards (Kosmopoulos et al., 2015)—we used tokenizer offset mappings to reconstruct character-level spans directly. Finally, lightweight post-processing rules were applied to merge fragmented spans, trim punctuation, and refine specific label predictions.

4 Experimental Setup

We used a stratified 10% validation split by genre for checkpoint selection. The model was trained for 12 epochs using AdamW, mixed precision, batch size 8, and gradient accumulation of 2. The learning rates were set to 3×10^{-5} for the encoder and 1×10^{-4} for the tagging head.

5 Results

5.1 Official Test Results

Table 1 compares our system's performance on the hidden test set against the baseline. Our system substantially outperformed the baseline across both genres, improving the overall F1 score by 0.447 absolute points and ranking 4th in the closed track.

System	Editorial	Debate	Overall
Official Baseline	0.272	0.293	0.285
Our System	**0.659**	**0.761**	**0.732**

Table 1: System vs. baseline F1 scores on the Daleel 2026 Task 2 hidden test set.

5.2 Discussion

As shown in Table 1, the model was more effective at capturing argumentative structure in debate transcripts than in editorials. Editorials remained challenging due to denser evidence types and subtle distinctions between AS, AN, TE, and ST.

Key design decisions that drove performance included the AraBERT contextual encoder, the Conformer tagging head, and span-fragmentation post-processing. Specifically, merging fragmented AN spans notably improved the official score, revealing that many model errors stemmed from over-segmentation rather than failed detection. Furthermore, a final rule converting selected numeric/value-based editorial AN spans into ST provided additional score gains.

6 Conclusion

We presented a closed-track system for Daleel 2026 Task 2 that combines a pretrained AraBERT encoder with a Conformer tagging head to detect Arabic Argumentative Discourse Units. By integrating overlapping token windows, class-weighted training, character-level span reconstruction, and lightweight post-processing, our system achieved a 0.732 overall F1 score.

Our analysis highlights that while debate texts are modeled effectively, fine-grained evidence classification in editorials remains challenging. Furthermore, resolving over-segmentation via post-processing is a highly effective strategy for span-overlap metrics. System limitations include reliance on a single encoder and validation split, making heuristics sensitive to data distribution. Future work will explore model ensembling, stronger Arabic encoders, constrained decoding, and data augmentation to better differentiate semantically close evidence types and improve rare-class detection.

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